

MATH Assignment 1 — Matrices & Linear Algebra

Mathematics (MATH) · Unit 1 · Matrices & Linear Algebra · 2025

Answer all questions. Show working and cite references where used.

Questions

1. Explain Rank of a Matrix with a neat diagram and a suitable example.
2. What is meant by Systems of Linear Equations? Discuss its advantages and limitations.
3. Illustrate Eigenvalues and Eigenvectors with a worked example.
4. Define Cayley-Hamilton Theorem. Derive or justify the key properties involved.
5. Illustrate Diagonalization with a worked example.
6. Discuss the role of Quadratic Forms in Matrices & Linear Algebra.